# RANS Simulation of a Swirled Premixed Flame in a Combustor Injector

Design Project Report

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# Nomenclature

## Latin Letters

| | | |
|---|---|---|
| c | Progress Variable | - |
| $k$ | Turbulent kinetic energy | $m^2 \cdot s^{-2}$ |
| P | Pressure | atm |
| T | Temperature | K |
| t | time | s |
| $S_L$ | Laminar flame speed | m/s |
| $S_T$ | Turbulent flame speed | m/s |
| $X_{[.]}$ | Molar fraction | - |

## Acronyms / Abbreviations

| | |
|---|---|
| 0D, 1D, 2D, 3D | Zero-, One-, two-, three-dimensional |
| CRZ | Central Recirculation Zone |

## Greek Symbols

| | | |
|---|---|---|
| $\mu$ | Dynamic viscosity | $m^2$/s |
| $\rho$ | Fluid density | kg/ $m^3$ |
| $\epsilon$ | Turbulent dissipation rate | $m^2 \cdot s^{-3}$ |
| $\omega$ | Specific dissipation rate | $s^{-1}$ |
| $\mu_t$ | Turbulent (eddy) viscosity | Pa·s |
| $\sigma_k$ | Turbulent Prandtl number for $k$ | - |
| $\sigma_\epsilon$ | Turbulent Prandtl number for $\varepsilon$ | - |
| $\sigma_\omega$ | Turbulent Prandtl number for $\omega$ | - |
| $\nu$ | Kinematic viscosity ($\nu = \mu/\rho$) | $m^2 \cdot s^{-1}$ |
| $\nu_t$ | Turbulent kinematic viscosity | $m^2 \cdot s^{-1}$ |

# Introduction

This report presents a numerical study of the reactive flow in the injector of the combustor of the future SNECFRAN engine, focusing on the simulation of a turbulent swirled premixed flame. The main objective is to assess whether a stationary and physically consistent solution can be obtained for the prescribed operating point, given the provided geometry, the selected turbulence model, and a premixed combustion model based on a progress-variable equation.

The configuration is characterized by a fully premixed fuel–oxidizer mixture, a high axial inlet velocity, a significant azimuthal component generating a swirling flow, and prescribed turbulence levels through the intensity and integral length scale. In this context, the flow structure, the possible formation of internal recirculation zones, and the interaction between turbulence and the flame front play a key role in flame stabilization and in the resulting temperature distribution inside the combustor.

From a modeling standpoint, the study relies on a steady Reynolds-Averaged Navier–Stokes (RANS) approach coupled with the two-equation $k$-$\epsilon$ turbulence model, which is widely used for jets and shear turbulent flows in reacting configurations. Premixed combustion is described using a $c$-equation type model, in which the chemistry is reduced to a transported scalar progress variable associated with a prescribed adiabatic flame temperature and laminar flame speed. This approach allows the position and topology of the flame front to be captured at a reasonable computational cost, while accounting for the influence of turbulence on flame propagation.

The work is structured into three complementary parts. First, the numerical case is set up in ANSYS Fluent, detailing the mesh, the physical models, the boundary conditions, and the numerical settings, and a first premixed reactive simulation is performed and analyzed through velocity, temperature, progress-variable, and turbulent kinetic energy fields. The second part is devoted to a parametric analysis of the influence of the critical stretch rate at extinction on the existence of stationary solutions, and to a comparison with a theoretical reference value. Finally, the third part discusses the limitations of the chosen models with respect to the real physics of a combustor injector and proposes possible improvements, while exploring, where feasible, enhanced simulations on the same mesh.

# 1 Part 1: Setup and initial calculation

## 1.1 Methodology

The study consists of performing a simulation using Reynolds Average Navier Stokes incompressible without spray of a swirl premixed flame under ANSYS Fluent. The calculation is based on the steady-state solution of the following system of equations (7 equations in total):

**Continuity Equation**

$$\nabla \cdot (\rho u) = 0 \tag{1}$$

**Momentum Conservation Equation**

$$\nabla \cdot (\rho u u) = -\nabla p + \nabla \cdot \left[\mu(\nabla u + \nabla u^T) - \rho u' u'\right] \tag{2}$$

where the Reynolds tensor $\rho u' u'$ is modeled using turbulent viscosity $\mu_t$.

**Turbulence model** 2 equations for each model explained in the appendix

**Premixed combustion model (c-equation)** The progress variable $c$ is defined such that $c = 0$ corresponds to unburnt gases and $c = 1$ to fully burnt products. Its transport equation reads:

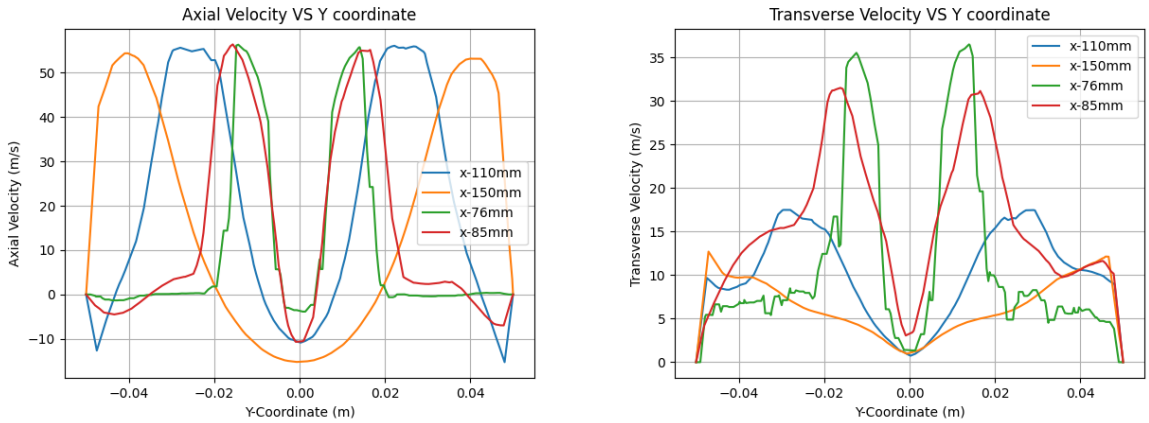
$$\nabla \cdot (\rho c \mathbf{u}) = \nabla \cdot (\Gamma_c \nabla c) + \dot{\omega}_c, \tag{3}$$

where $\Gamma_c = \mu + \dfrac{\mu_t}{Sc_t}$ is the effective diffusivity (molecular plus turbulent) and $\dot{\omega}_c$ is the combustion source term, often modeled in terms of a (laminar or turbulent) flame speed.

## 1.2 Flow field and flame structure analysis

### 1.2.1 Velocity field and flow structure

The profiles of axial and tangential velocity provide complementary information on the organization of the swirling premixed flow inside the combustor. Although the exact quantities are plotted at different axial positions, the profiles correspond to three distinct regions that control flame stabilization and global flame topology.




(a) Axial Velocity

(b) Tangential Velocity

Figure 1: Profiles visualization of the axial and tangential velocity

In the upstream region close to the injector exit, the axial velocity profiles remain close to the imposed inlet conditions, and the flow is essentially unaffected by combustion. This confirms the correct implementation of the boundary conditions and corresponds to the fresh premixed gases before significant heat release.

Further downstream, the profiles clearly show the development of a central recirculation zone, with reduced or even negative axial velocity along the centerline and strong tangential motion in the shear layers. This recirculation, induced by the imposed swirl, transports hot burned products upstream and creates a region of high residence time that stabilizes the premixed flame and prevents blow-off.

In the fully developed combustion region, the axial velocity profiles exhibit a marked acceleration of the flow, together with off-axis peaks associated with intense shear layers and swirl-induced mixing. This increase in axial velocity is primarily due to the thermal expansion of the burned gases after heat release, and it is consistent with the observed shape and position of the turbulent flame front along the combustor.

Figure 1b shows, at different axial positions, tangential velocity profiles that are almost symmetrical about the axis: the velocity remains low at the center, reaches a maximum in two lateral shear zones, then decreases to nearly zero at the edges, corresponding to a twisted jet surrounded by two shear layers. In the immediate vicinity of the injector outlet, the peaks are higher and more pronounced, indicating a more intense swirl and stronger fluid rotation, whereas further away along the chamber, these peaks decrease and the curves become flatter, reflecting the gradual diffusion of the swirl momentum and the decrease in swirl with axial distance.

### 1.2.2 Static temperature field

The static temperature distribution provides direct information on the heat release and on the position of the flame front, and it can be used to assess the validity of the adiabatic assumption adopted in the premixed combustion model.

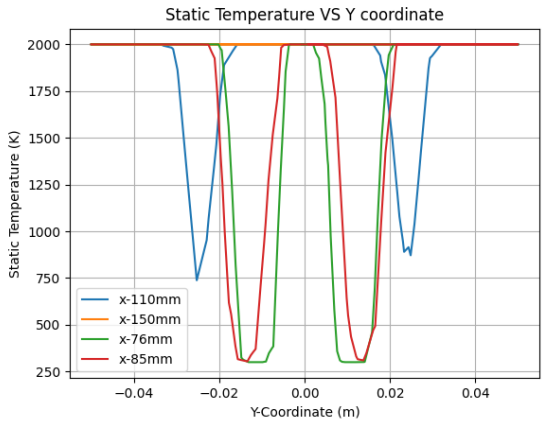



Figure 2: Transverse profiles of static temperature at different axial locations inside the combustion chamber.

The temperature profiles exhibit a rapid rise from the fresh mixture temperature to values close to the adiabatic flame temperature, with sharp gradients that clearly delineate the flame front. Further downstream, mixing between hot burnt gases and colder surrounding fluid leads to a more uniform temperature field, indicating progressive homogenization of the thermal field in the combustor core.

### 1.2.3 Progress variable and flame topology

The progress variable $c$, obtained from the premixed combustion model based on the *c-equation*, provides a convenient scalar to identify the flame structure and topology in the combustor.

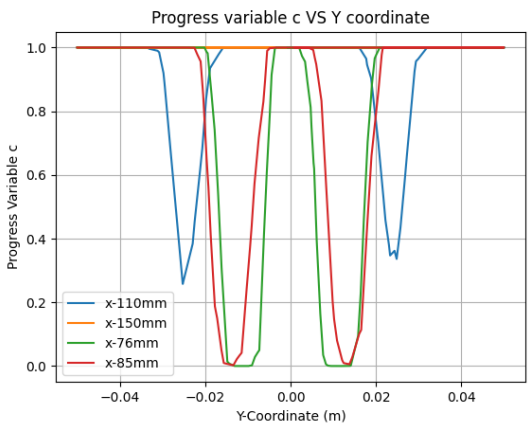


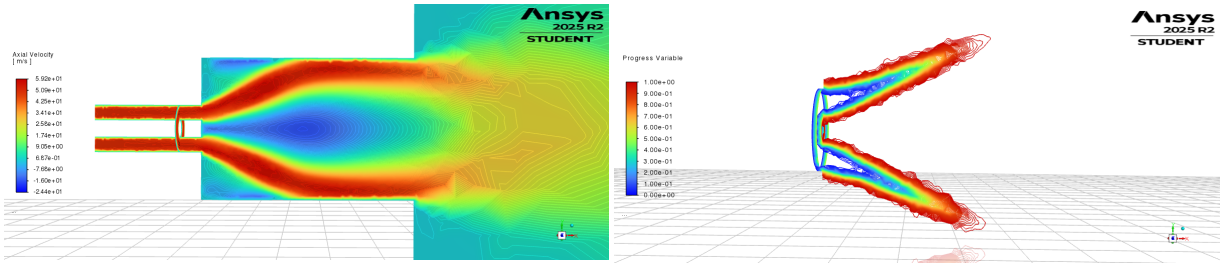

Figure 3: Transverse profiles of the progress variable $c$ at different axial positions.

Values of $c \approx 0$ correspond to fresh premixed gases, whereas values close to unity indicate fully burnt products, so the region where $0 < c < 1$ clearly marks the reaction zone and the instantaneous posifresh-mixturelame front. The axial enear of these profiles reveals how the flame surface is anchored in the vicinity of the central recirculation zone and how it expands downstream, confirming that the adopted turbulence–combustion model is able to sustain

a stable premixed flame with the expected V-shaped topology under the present operating conditions.

### 1.2.4 Flame stabilization mechanism and topology

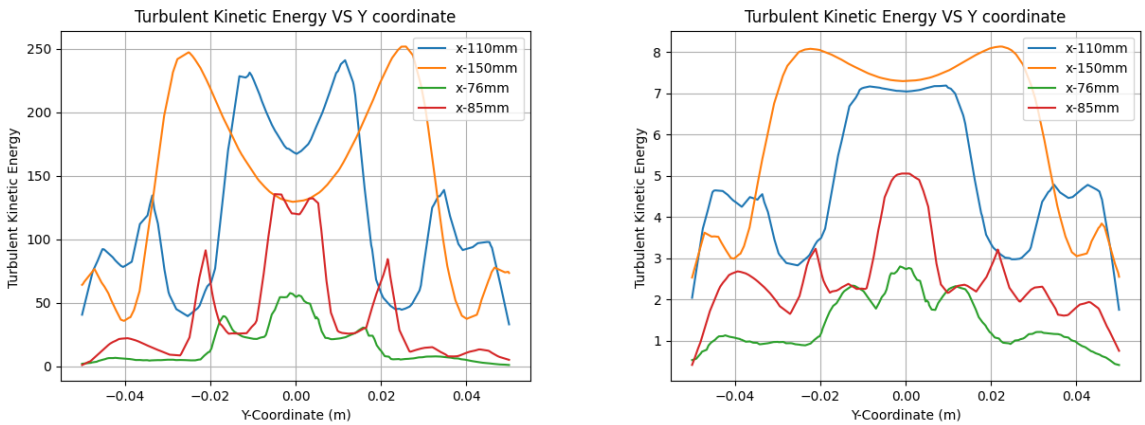
To define the mechanisms that enable flame stabilization, we must first understand what constitutes a stable flame. In a premixed flow, the flame is stable when it maintains a stationary or quasi-stationary average position despite the presence of strong turbulence. This stability is due to the presence of a state of equilibrium between the turbulent flame speed and the normal gas velocity.




(a) Contours of Axial Velocity of Progress Variable

aptionComparison between the axial velocity and the progress variable canofStabilization

In this geometry, the swirling flow generates a central recirculation zone (CRZ) downstream of the injector, linked to the jet's high kinetic momentum.




(a) Turbulent Kinetic Energy (b) Turbulent Flame Speed

Figure 5: Visualization of turbulent kinetic energy and the turbulent flame speed profiles

Based on the axial velocity and turbulent kinetic energy profiles, this recirculation zone is located in the center of the combustion chamber, where the axial velocity becomes almost zero or slightly negative, while k reaches high values. The burned gases returning upstream preheat the fresh mixture, ensuring continuous ignition and maintaining a stable flame over time. The turbulent flame velocity profile shows that the flame is at its maximum near the flame front, where the turbulence is most intense.

Three main mechanisms are associated with CRZ:

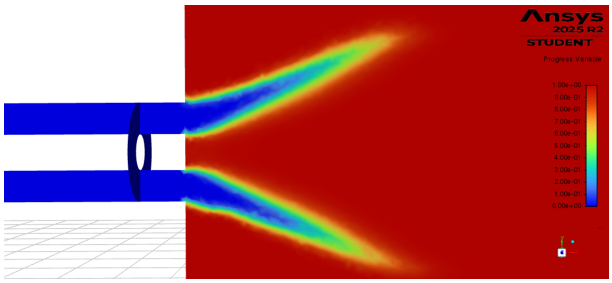
- Preheating of the fresh mixture by recirculated combustion gases in the central recirculation zone raises the local temperature above the auto-ignition temperature.
- A decrease in axial velocity in this zone, linked to the return movement of the gases, which increases the residence time and allows the preheated mixture to ignite and remain in contact with the flame.
- A local increase in turbulence, which increases the turbulent flame velocity and, combined with preheating, maintains quasi-continuous ignition of the recirculated fresh mixture, thus ensuring flame stability.

In terms of topology, the conditions of our simulation lead to a V-shaped flame, typical of twisted premixed flames stabilized by a central recirculation zone. The shape of the flame depends mainly on the operating parameters, in particular the stabilization criterion chosen (notably the critical value of g), which determines the balance between the turbulent flame velocity and the normal gas velocity, as discussed previously in the definition of a stable flame.

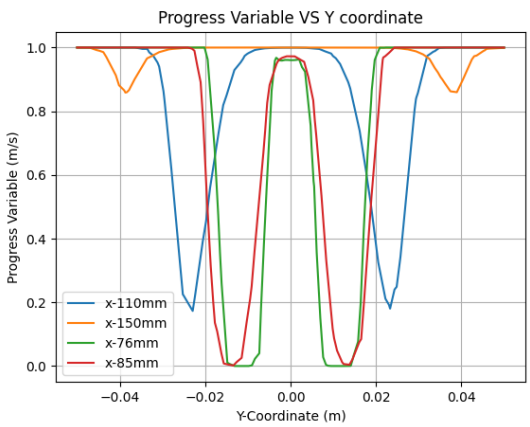
# 2 Part 2: Analysis of the effect of local flame extinction by stretching

## 2.1 Parametric study on stretching extinction

To investigate the influence of the critical strain rate for extinction on the premixed flame, three reactive simulations were performed with different values of the critical stretching factor $g_{cr}$. Starting from the default value prescribed in Fluent ($1.0 \times 10^8\ \mathrm{s}^{-1}$), two additional cases were computed by progressively reducing this parameter toward the theoretical critical strain rate of $2.0 \times 10^4\ \mathrm{s}^{-1}$.



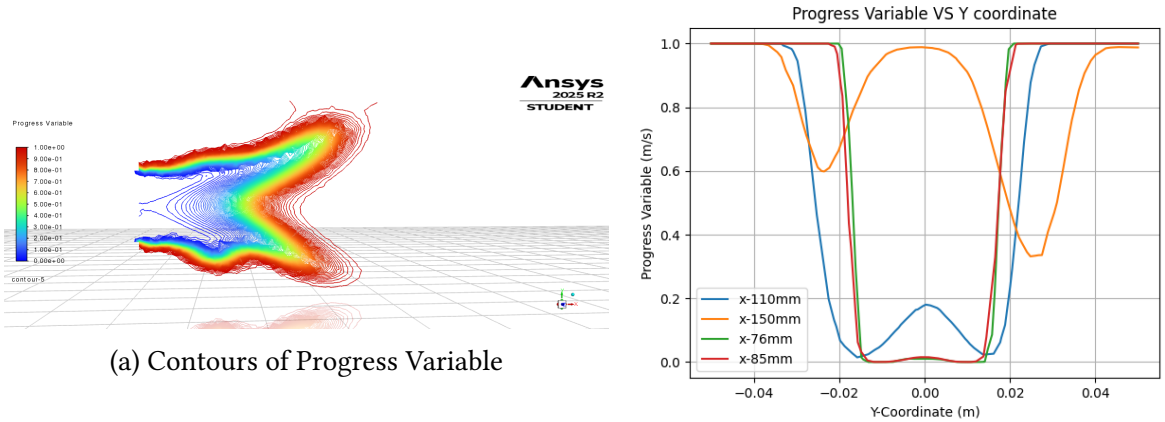

(a) Contours of Progress Variable




(b) Progress Variable profile

Figure 6: Visualization of the contours and profile of the progression variable for $g_{cr} = 2.10^4$

The transverse fields of progress variable and temperature show that decreasing the critical strain rate leads to a stronger stretching of the flame front and to a noticeable thickening of the reaction zone. For moderate reductions of $g_{cr}$, a statistically stationary solution is still obtained, but the flame surface becomes more wrinkled and extends further into the recirculation region, indicating an increased sensitivity of the combustion process to local strain effects.




(a) Contours of Progress Variable

(b) Progress Variable profile

Figure 7: Visualization of the contours and profile of the progression variable for $g_{cr} = 2.10^4$

When $g_{cr}$ is decreased below an order of magnitude comparable to the theoretical value of $2.0 \times 10^4\ \mathrm{s}^{-1}$ (see Figure 7), the solver no longer converges to a fully stationary state and large unsteady pockets of locally extinguished flame appear. In this regime, the flame topology changes from a classical V-shape to an M-shape, with partial quenching near the injector lip and only isolated branches of the flame surviving in the shear layers, showing that excessive stretching no longer promotes flame stabilization but can instead trigger local extinction.

# 3 Part 3: Areas for improvement and discussion

The present simulation provides a first insight into the global flow organization and flame stabilization mechanisms in the premixed swirling combustor, but several important physical phenomena are either neglected or only crudely represented. This section identifies the main limitations of the current setup and suggests possible improvements to bring the configuration closer to a realistic injector while keeping the same mesh.

## 3.1 Physical modelling limitations

First, the premixed turbulent combustion is described with the adiabatic *c-equation* model, which assumes a thin flame front and neglects detailed finite-rate chemistry and heat losses. In a real injector, non-adiabatic effects due to wall heat transfer, radiation, and incomplete combustion may significantly affect flame temperature, laminar flame speed, and local extinction behavior.

Second, the turbulence is modelled with a two-equation RANS $k$–$\varepsilon$ formulation, which is known to be only partially adequate for strongly swirling, anisotropic and recirculating flows typically found in swirl-stabilized combustors. The model cannot correctly reproduce the turbulence anisotropy and may mispredict the size and strength of the central recirculation zone and the associated shear layers.

Third, the inlet conditions are idealized: the mixture is assumed to be perfectly premixed, with uniform velocity and turbulence profiles, and without any geometric details of the injector (vanes, swirler blades, fuel injection holes, etc.). In practice, imperfect premixing, velocity profile distortions and small-scale injector details strongly influence mixture fraction stratification, local equivalence ratio and flame anchoring.

## 3.2 Numerical and boundary-condition aspects

The present computation is performed at steady state using a RANS approach, which filters out the inherent unsteadiness of swirl-induced instabilities and turbulent flame dynamics. Important phenomena such as precessing vortex core motion, large-scale vortex shedding, and intermittent local extinction–reignition events are therefore not captured.

In addition, the mesh has been generated for a first design study and does not result from a systematic grid-sensitivity analysis. Even if it is kept unchanged in the present work, its resolution in the shear layers, near-wall regions and in the flame zone may be insufficient to accurately represent gradients of velocity, temperature, progress variable and turbulent kinetic energy. The treatment of wall boundary conditions is also simplified, since realistic wall temperatures or conjugate heat transfer are not included.

## 3.3 Proposed improvements toward a more realistic injector

A first straightforward improvement would be to enrich the combustion model by enabling non-adiabatic effects and using tabulated laminar flame speeds that depend on mixture composition and unburnt gas temperature. This would allow a more realistic prediction of flame temperature, local burning velocity and extinction limits, especially in regions affected by wall heat losses or mixture inhomogeneities.

For the turbulent flow, a more advanced turbulence model better suited to swirling recirculating flows, such as the realizable or RNG $k$–$\varepsilon$ model, the SST $k$–$\omega$ model, or even a Reynolds Stress Model (RSM), could be employed without modifying the mesh. These models are known to improve the prediction of anisotropic turbulence and of the structure of the internal recirculation zone in swirl-stabilized combustors.[web:30][web:31][web:34]

The inlet and boundary conditions could also be refined by imposing more realistic axial and tangential velocity profiles, turbulence intensity and length scales consistent with experimental data or detailed injector design. Introducing partial premixing or stratified mixture effects, while keeping the same grid, would make it possible to study the sensitivity of the flame to local equivalence ratio variations and to better reproduce real operating conditions.[web:27][web:32]

## 3.4 Bonus: practical improvements on the existing setup

Without changing the mesh, several practical modifications can be tested to improve the present simulation. A first step is to perform a mesh-independence verification on the existing grid by locally refining the regions of high gradients (shear layers, flame front, near-wall zones) and comparing key quantities such as recirculation length, maximum temperature and turbulent kinetic energy to ensure that the base mesh is sufficiently resolved.

Then, a series of additional simulations can be carried out by switching from the standard $k$–$\varepsilon$ model to a realizable or RNG $k$–$\varepsilon$ model and by enabling non-adiabatic premixed combustion options in the *c-equation* framework. Comparing axial and radial velocity profiles, flame shape (V- or M-shaped), and integral quantities such as pressure drop and heat release between these cases and the reference solution would show whether the modified model better captures the expected behaviour of a real injector. Finally, if computational resources allow, an unsteady RANS calculation on the same mesh could be performed to assess the impact of large-scale flow unsteadiness on flame stabilization and local extinction.

# Conclusion

The numerical study of the premixed swirling combustor has highlighted the strong coupling between aerodynamics, turbulence and combustion in a swirl-stabilized injector. The reference simulation reproduces the expected flow organization, with a central recirculation zone generated by swirl, a V-shaped premixed flame, and a clear correlation between regions of low axial velocity, high turbulent kinetic energy and intense heat release.

The analysis of the static temperature and progress-variable fields confirms that the adiabatic premixed combustion model based on the c-equation is able to capture a globally stable flame and its anchoring mechanism, while also revealing the strong thermal gradients and reaction zones that control the overall flame topology. The parametric study on the critical strain rate for extinction shows that decreasing the extinction criterion increases flame stretching and thickness, and that approaching the theoretical critical value leads to a transition from a V-shaped to an M-shaped flame with large regions of local quenching, beyond which a fully stationary solution can no longer be obtained.

At the same time, the work has underlined several limitations of the current modelling strategy, in particular the use of a RANS $k$–$\varepsilon$ turbulence model, idealized inlet conditions and an adiabatic premixed flame representation. The proposed improvements—more advanced turbulence closures, non-adiabatic premixed combustion modelling, refined boundary conditions and, eventually, unsteady simulations—constitute natural next steps to bring the configuration closer to a realistic injector while keeping the same mesh and to increase the predictive capability of the numerical tool for future design studies.

# Appendix

## Comparison between $k-\epsilon$ and $k-\omega$ SST turbulent model

- **$k$-$\epsilon$ model**

$$\nabla \cdot (\rho k \mathbf{u}) = \nabla \cdot \left[ \left( \mu + \frac{\mu_t}{\sigma_k} \right) \nabla k \right] + P_k - \rho \epsilon \tag{4}$$

$$\nabla \cdot (\rho \epsilon \mathbf{u}) = \nabla \cdot \left[ \left( \mu + \frac{\mu_t}{\sigma_\epsilon} \right) \nabla \epsilon \right] + C_{1\epsilon} \frac{\epsilon}{k} P_k - C_{2\epsilon} \rho \frac{\epsilon^2}{k} \tag{5}$$

$$\mu_t = C_\mu \rho \frac{k^2}{\epsilon}$$

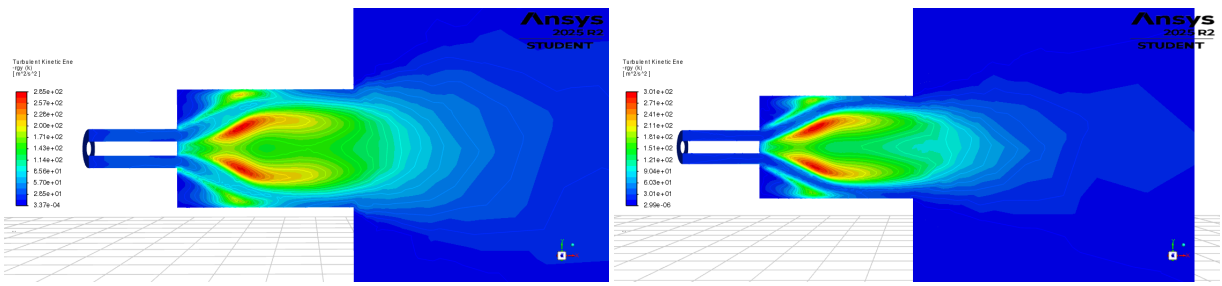
- **$k$-$\omega$ SST model**

$$\nabla \cdot (\rho k \mathbf{u}) = \nabla \cdot \left[ \left( \mu + \frac{\mu_t}{\sigma_k} \right) \nabla k \right] + P_k - \beta^* \rho k \omega \tag{6}$$

$$\nabla \cdot (\rho \omega \mathbf{u}) = \nabla \cdot \left[ \left( \mu + \frac{\mu_t}{\sigma_\omega} \right) \nabla \omega \right] + \alpha \frac{\omega}{k} P_k - \beta \rho \omega^2 \tag{7}$$

$$\mu_t = \frac{\rho k}{\omega} \qquad \text{(basic formulation, before SST limiting)}$$

### Turbulent kinetic energy

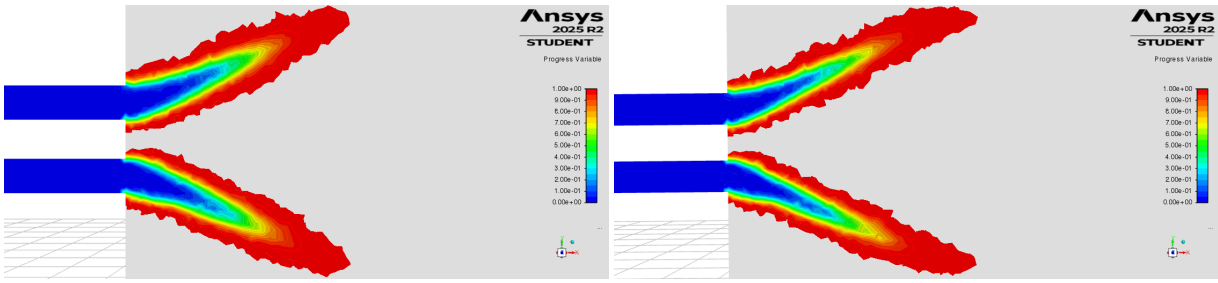



(a) C$k-\epsilon$ turbulent model

(b) $k-\omega$ SST turbulent model

Figure 8: Comparison of the turbulent kinetic energy contours of the two models

## Progress variable




(a) $k - \epsilon$ turbulent model (b) $k - \omega$ SST turbulent model

Figure 9: Comparison oh the Progress variable contours of the two turbulent models

The figure compares the two turbulence models used in this study based on turbulent kinetic energy contours. We can see that turbulence levels are significantly higher and more diffuse with the k-$\epsilon$ model than with the k-$\omega$ SST model.

Based on the concepts covered in the turbulence course, we know that the k-$\epsilon$ model is particularly suited to flows far from walls and relies on wall laws to model the boundary layer, which limits its accuracy when near-wall effects become significant. Conversely, the k-$\omega$ SST model combines the k-$\omega$ model near the wall with the k-$\epsilon$ model in the free stream, thereby improving the description of the boundary layer and shear zones.

According to the course, the k-$\omega$ model tends to overestimate turbulence in highly sheared regions, leading to over-diffusion of the mixture and, consequently, a larger turbulent kinetic energy zone in our results. The k-$\omega$ SST model yields more moderate and better-localized turbulence levels, leading to a more reliable estimate in configurations where wall and recirculation effects are not negligible.

It would be interesting to add an antisymmetric term to better capture vortices, as this is not possible in the k-$\epsilon$ model due to its symmetric term, which does not account for rotational phenomena.